

MAKER-X

SCHOOL OF CREATIVITY & ART · SIST · SUMMER 2026

You don't just **build** a robot — you conceive the idea, design its personality, build the electronics, and then **pitch it**. Four weeks. One team. One product you own end-to-end.

你不只是在制造机器人——你要构思创意、设计个性、搭建电路，最终登台展示。四周时间，一支团队，一件完全属于你的作品。

4

WEEKS

12h

PER WEEK

¥600PER TEAM
BUDGET**2-3**STUDENTS /
TEAM**Pass / No
Pass**

GRADING

// THE MISSION

Conceive it. Design it. Build it. Pitch it.

Maker-X is an intensive **4-week project course** where cross-disciplinary teams build AI-powered robotic companions from scratch — hardware, firmware, personality, and pitch. Instruction in **Chinese & English**.

01

**Conceive**

Identify a real user and a genuine need. Define the personality and the purpose of your robot before touching any hardware.

02

**Design**

Shape, sound, language, expression — every detail of the robot's soul. Prototype rapidly and iterate without fear.

03

**Build**

Solder, code, 3D-print, and integrate. Electronics, firmware, enclosure — one complete, working product.

04

**Pitch It**

Present your product to real university guests. A working demo, a poster, a story — your public debut.

// COURSE OBJECTIVES

What you'll walk away with



Cross-discipline Collaboration

Work in teams that span engineering, design, and beyond. Communicate across disciplines and ship something greater than the sum of its parts.



Confidence with AI & LLM Tools

Integrate large language models into real hardware — learning how to prompt, tune, and deploy AI as a core part of a product.



Design Thinking in Practice

Empathy, creativity, rapid prototyping, iteration, and communication — not as theory, but as daily practice.

// CORE TECHNOLOGY

What you'll build with



Hardware

ESP32-based boards as the brain — Wi-Fi, Bluetooth, real-time AI interaction.

Displays, sensors, cameras, and motors for awareness and expression. Body **3D-modeled and printed** — designed by you, from scratch.



Frameworks

ESP-Claw — firmware framework for ESP32 AI robots, handling voice, display, and LLM integration.

OpenRouter — unified API gateway to switch between any LLM with a single interface.



Large Language Models

State-of-the-art LLMs powering your robot's intelligence:

Qwen · DeepSeek · Claude ...and others.

// MISSION TIMELINE

Four weeks. One product.

The course runs at **12 hours per week** — enough time to go deep, not enough time to hesitate.

WEEK ONE

Conceive

User research, concept selection, personality brief

WEEK TWO

Design

Prototyping, firmware, form & character development

WEEK THREE

Build

User testing, refinement, 3D-print the enclosure

WEEK FOUR ★

Pitch It

Demo & pitch to university guests — poster presentation

// INSTRUCTORS

Meet your team

A joint offering from **Kedao College**, **SCA** (School of Creativity and Art), and **SIST** (School of Information Science and Technology).



INSTRUCTOR
Shimon Shmueli
SCA



INSTRUCTOR
Daniel Koo
SCA



INSTRUCTOR
梁俊睿
SIST

Ready to build something real?

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ShanghaiTech University
Kedao College · SCA · SIST

